

Hiver SDE Intern — AI Customer Support Agent

Brand: AmazonHelp **Dataset:** Customer Support on Twitter

1. Problem Framing

The goal is to build a support agent that classifies incoming customer messages, retrieves historically similar AmazonHelp resolutions, drafts a grounded response, and decides whether the case can be auto-handled or should be escalated.

2. Data and Intent Taxonomy

AmazonHelp was selected because it provides a large volume and wide variety of customer-support interactions. Direct customer messages followed by AmazonHelp responses were constructed as historical resolution pairs. After cleaning and deduplication, 154,910 pairs were available for retrieval.

The taxonomy contains 14 intents: `delivery_not_received`, `delivery_delayed`, `delivery_failed_or_wrong`, `order_status`, `return_or_refund`, `payment_or_billing`, `account_access`, `prime_subscription`, `prime_video`, `product_or_device`, `product_information`, `customer_service_complaint`, `fraud_or_security`, and `other`.

3. Golden Evaluation Set

A 200-example golden set was sampled across discovered issue clusters to increase coverage of diverse language and issue patterns rather than estimate the natural class distribution. The final set contains 200 labeled examples.

Important limitation: `prime_video` has zero examples in the final golden set and `fraud_or_security` has only two. Therefore, per-class results for sparse intents should not be treated as reliable production estimates.

4. Baselines and Classification Results

System	Accuracy	Macro-F1
Majority-class baseline	20.5%	2.6%
TF-IDF + Logistic Regression	43.0%	34.8%

The TF-IDF + Logistic Regression classifier was evaluated using out-of-fold predictions on the 200-example golden set. This prevents the model from being evaluated on examples used to fit the corresponding fold.

5. Historical Retrieval and Response Drafting

Historical customer messages are represented using TF-IDF unigrams and bigrams. Cosine similarity retrieves similar customer cases. Very short, generic, or unsuitable historical responses are filtered before being presented as evidence.

The current prototype uses intent-specific response templates for drafting. Retrieved historical cases provide evidence for how similar AmazonHelp issues were historically handled.

6. Auto-Handle vs Escalate

High-risk intents such as `fraud_or_security` and `customer_service_complaint` are escalated. The system also escalates when classifier confidence or historical retrieval evidence is insufficient.

Observed result: under the conservative 0.50 confidence and 0.50 retrieval thresholds, 0/200 golden examples were auto-handled and 200/200 were escalated. This is treated as a safety finding rather than lowering thresholds merely to increase automation.

7. Top Failure Modes

Delivery vs Prime confusion: A delayed delivery mentioning Amazon Prime was classified as `prime_subscription`.

Delivered vs delayed confusion: Messages stating that tracking says delivered but the customer did not receive the item can be classified as `delivery_delayed`.

Billing vs other confusion: A billing-address question in French was classified as `other`.

Product information vs billing confusion: A question about the Amazon Brazil pre-sale page was classified as `payment_or_billing`.

Retrieval similarity is not sufficient: A highly similar historical message can still have a generic or unsuitable resolution, so retrieval results require response-quality filtering.

8. What Is Misleading About My Headline Number?

The 43.0% accuracy / 34.8% macro-F1 result is measured on only 200 examples selected for coverage rather than natural class frequency. Several intents have very small support and `prime_video` is absent. Therefore the number should not be interpreted as production accuracy. The OOF protocol makes the estimate less optimistic than an in-sample evaluation, but the small and non-representative test set remains a major limitation.

9. Non-Obvious Design Decisions

1. Selected AmazonHelp because of its large interaction volume.
2. Constructed direct customer-to-brand response pairs.
3. Used a sampled retrieval corpus rather than the entire dataset for practical runtime.
4. Used TF-IDF word unigrams/bigrams as a transparent retrieval baseline.
5. Filtered low-value historical responses before using them as evidence.
6. Defined a compact 14-intent taxonomy from observed issue patterns.
7. Used cluster-balanced sampling for golden-set coverage.
8. Used out-of-fold predictions to prevent evaluation leakage.
9. Kept the majority classifier as a deliberately weak baseline.
10. Used class-balanced Logistic Regression for the simple learned baseline.
11. Kept high-risk intents on mandatory escalation.
12. Used conservative evidence gates for autonomous handling.
13. Did not lower thresholds merely to manufacture auto-handled examples.
14. Treats classifier confidence as an evidence score rather than calibrated probability.

10. Next Week

The next iteration would improve semantic intent classification, calibrate confidence, use stronger semantic retrieval/reranking, expand the golden set for sparse intents, and evaluate automation coverage and precision on a larger representative test set. Response generation can also be upgraded to an LLM constrained by retrieved historical resolutions.

11. Evaluation Note

The evaluation harness includes automated intent metrics and response-quality evaluation components. LLM-as-judge and human agreement results should only be reported with the measured scores from the completed judge/human evaluation; no agreement value is fabricated here.